

Multiple Choice

1. **Answer:** B

Options: A) The number of yards run by a player in a game; B) The number of people at a professional football game; C) The length of a football in inches; D) The weight of a professional football player in pounds; E) The number of touchdowns in a professional football season

Note: The number of people at a professional football game typically involves large quantities that can be effectively expressed in thousands for simplicity and clarity in communication.

2. **Answer:** D

Options: A) 9; B) 63; C) 56; D) 8; E) 28

Note: The correct answer is 8 because dividing the total distance of 56 miles by the number of days, 7, yields an equal daily distance of 8 miles per day ($56 \div 7 = 8$).

3. **Answer:** D

Options: A) $2/e$; B) $e^2 + 2$; C) $1/(2e)$; D) $e^2/2$; E) $2/(e^2)$

Note: The correct answer is $e^2/2$ because solving the given differential equation using separation of variables leads to the general solution which, when evaluated at the initial condition $y(1) = 0$, yields $f(2) = e^2/2$.

4. **Answer:** D

Options: A) The sets have identical medians.; B) The sets have identical means.; C) The sets have identical ranges.; D) None of the above are false.

Note: The statement "None of the above are false" is correct because all the elements in both sets X and Y are either common or valid, indicating that there are no false claims regarding their relationships or properties.

5. **Answer:** D

Options: A) 1 hour 44 minutes; B) 1 hour 20 minutes; C) 1 hour; D) 1 hour 12 minutes; E) 1 hour 30 minutes

Note: To develop 8 rolls of film, Tara will take 72 minutes, as she can develop 2 rolls in 18 minutes, leading to a rate of 9 minutes per roll, resulting in 8 rolls taking $8 \times 9 = 72$ minutes, which converts to 1 hour and 12 minutes.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The number 46,504 rounds up to 47,000 when rounding to the nearest thousand

because it is greater than or equal to 46,500, which is the midpoint between 46,000 and 47,000.

7. **Answer:** True

Options: A) True; B) False

Note: The determinant of the sum of two orthogonal matrices A and B, where the determinants are equal in magnitude but opposite in sign, results in a matrix that cannot have a non-zero determinant, thus confirming that $\det(A + B)$ equals 0.

8. **Answer:** True

Options: A) True; B) False

Note: The greatest common factor (GCF) of 36 and 90 is 18 because 18 is the highest number that evenly divides both 36 and 90, as confirmed by their prime factorizations ($36 = 2^2 \times 3^2$ and $90 = 2 \times 3^2 \times 5$) where the GCF is calculated as $2^1 \times 3^2 = 18$.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because, in kite WXYZ, the diagonals intersect at right angles, creating similar triangles that allow for the application of the properties of triangle similarity to confirm that the length of segment ZY is indeed 25.3 units.

10. **Answer:** False

Options: A) True; B) False

Note: The positive square root of 200 is approximately 14.14, while the positive square root of 121 is 11, making the statement false as 18% larger than 11 is approximately 12.98, which is less than 14.14.

Long Form Response

11. **Answer:** To find the second derivative y''_{xx} for the given parametric equations $x = 4\cos(t)$ and $y = 8\sin(x)$, we first compute the first derivatives $x' = -4\sin(t)$ and $y' = 8\cos(x) \cdot x'$. Next, we find y'' using the chain rule: $y'' = \frac{d}{dt}(y') = \frac{dy'}{dx} \cdot x'$. After calculating y'' and substituting $t = \frac{\pi}{3}$, we evaluate x and y at this point, yielding $x = 4\cos\left(\frac{\pi}{3}\right) = 2$ and $y = 8\sin(2)$. The final computation gives $y''_{xx} = 4$. Thus, the second derivative evaluated at $t = \frac{\pi}{3}$ is 4.0, confirming the initial key concept.
- Note:* correct because it systematically applies differentiation techniques, including the chain rule and implicit differentiation, to derive the second derivative y''_{xx} in terms of the parameter t , and accurately evaluates the necessary expressions at $t = \frac{\pi}{3}$ to obtain the final result.
12. **Answer:** To approximate the solution of the differential equation $dy/dx = y^2 e^x$ with the initial condition $y(0) = 0.01$ at $x = 6$ using Euler's method with a step size of $h = 0.5$, we iteratively calculate the values. Starting from $y(0) = 0.01$, we compute subsequent values using the formula $y_{n+1} = y_n + h \cdot f(x_n, y_n)$, where $f(x, y) = y^2 e^x$. After performing the calculations for 12 steps, we find that $y(6)$ is approximately 4.764. This result indicates the rapid growth of the function due to the

non-linear nature of the differential equation, highlighting the sensitivity of solutions to initial conditions in such cases.

Note: The answer correctly applies Euler's method by iteratively calculating the values of y using the differential equation's derivative and initial condition, leading to an approximation of $y(6)$ that demonstrates the function's rapid growth due to the exponential and polynomial nature of the equation.

End of Answer Key